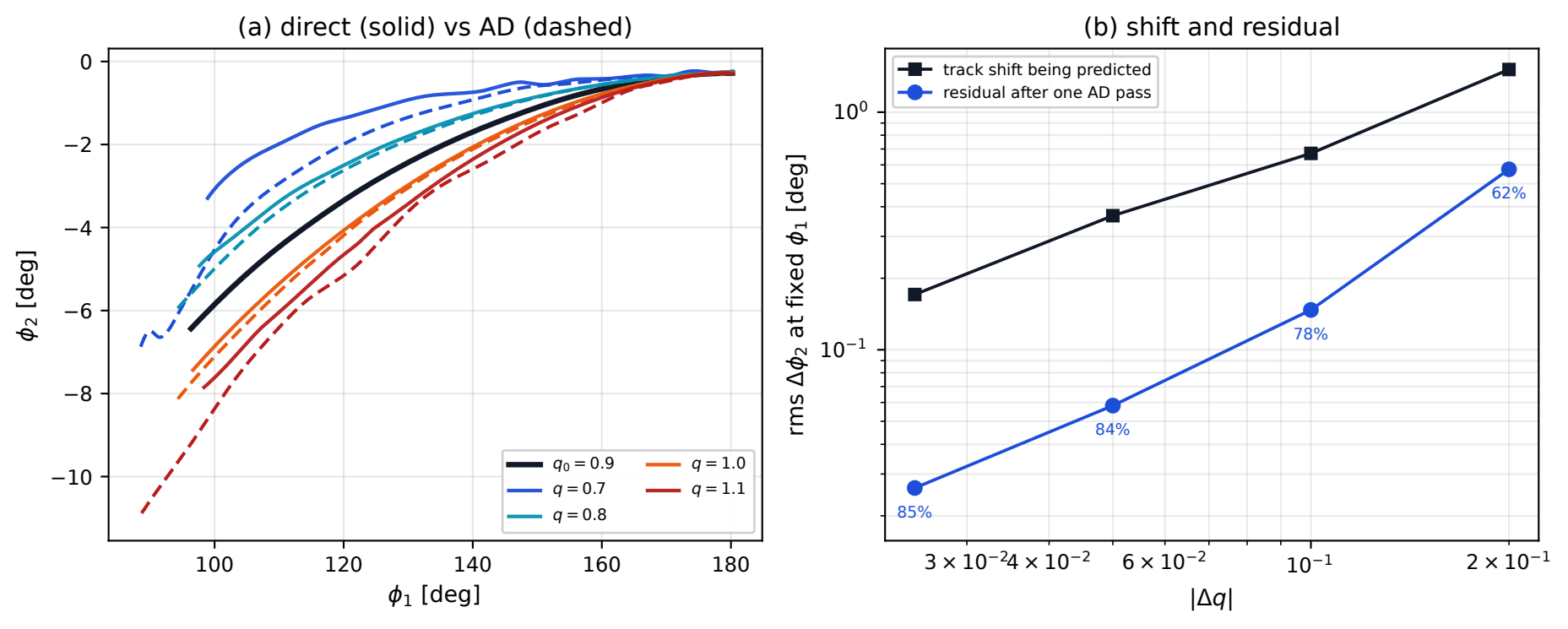
(a) direct (solid) vs AD (dashed)
ϕ_2 [deg]
ϕ_1 [deg]
0
−2
−4
−6
−8
−10
100
120
140
160
180
$q_0 = 0.9$
$q = 0.7$
$q = 0.8$
$q = 1.0$
$q = 1.1$
(b) shift and residual
rms $\Delta\phi_2$ at fixed ϕ_1 [deg]
$|\Delta q|$
track shift being predicted
residual after one AD pass
10^0
10^{-1}
3×10^{-2}
4×10^{-2}
6×10^{-2}
10^{-1}
2×10^{-1}
85%
84%
78%
62%